eCAMPUS UNIVERSITÀ | DiSTA

Lingua inglese

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DATA VISUALISATION

Types of numbers

- numbers in a group together may be called a **series** or **set of numbers**
- if the order in which they occur is significant they may be called a **sequence of numbers**
- 1, 3, 5, 7 … = **odd numbers**
- 2, 4, 6, 8 … = **even numbers**
- 2, 3, 5, 7, 11 … = **prime numbers**
- **approximate** number: one which is roughly correct but not the **precise** or **exact** number

DATA VISUALISATION

- **aggregate** number: a number reached by totalling a set of numbers = the **total**
- **discrete** number (unit): something which is separate and cannot be divided into smaller numbers (opposite: **continuous**)
- **constant** number: one that does not change

DATA VISUALISATION

Numbers are read in the following manner: million, thousand, hundred

e.g. 2,350,400 two million three hundred and fifty thousand four hundred

Decimals

2.36 two point three six

Percentages

37% thirty seven percent

Fractions

3/8 (three eights), ¼ (one quarter), 2/3 (two thirds), ½ (one half)

DATA VISUALISATION

Numbers and operations

+ plus (and)

- minus (take away)

x multiplied by (times)

÷ divided by

= equals (is)

. point

% percent

DATA VISUALISATION

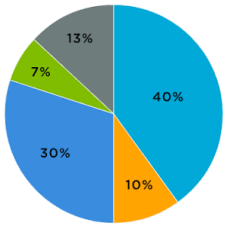
Diagrams

- visual ways of presenting data
- they are often called figures
- in an academic article they are usually labelled: Fig. 1, Fig. 2)

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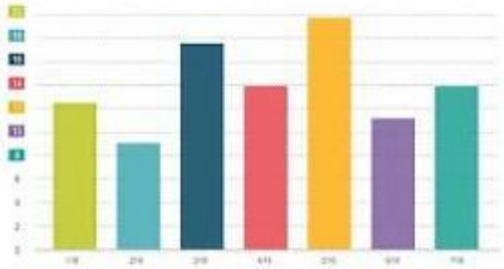
DATA VISUALISATION

Types of diagrams:



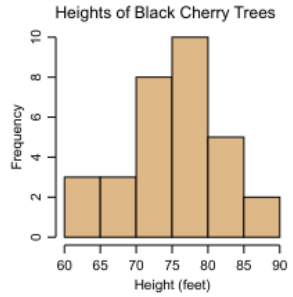
- **pie chart**: a circle divided into segments

- **bar chart**: different amounts are represented by thin horizontal or vertical lines

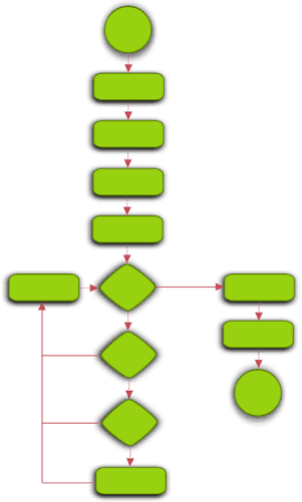


DATA VISUALISATION

- **histogram:** a bar chart; the bar width also varies to indicate different values

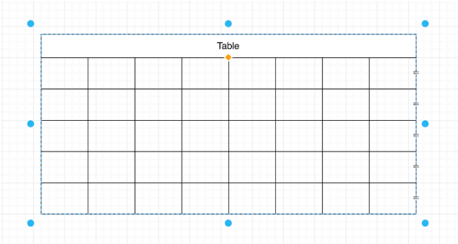


- **flowchart:** indicates the stages of a process

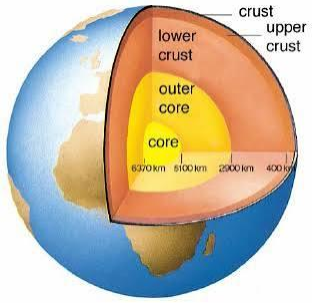


DATA VISUALISATION

- **table:** a grid with columns and rows



- **cross-section:** the model of something cut across the middle



DATA VISUALISATION

A graph:

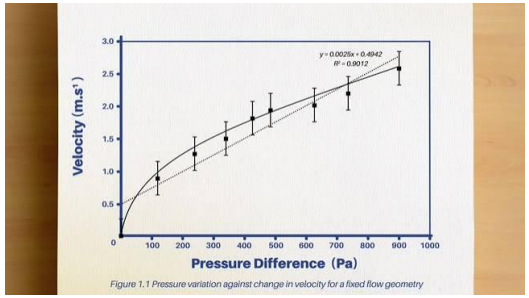


Figure 1.1 Pressure variation against change in velocity for a fixed flow geometry

- shows how different items or numbers relate to each other

x axis = horizontal axis

y axis = vertical axis

DATA VISUALISATION

- graphs show how numbers **increase** or **decrease**
- numbers can also be said to **rise** or **grow** and **fall**, **drop** or **decline**

Compare:

- NOUN: rise, growth, fall, drop, decline
- VERB: to rise, to grow, to fall, to drop, to decline
- INCREASE and DECREASE (different pronunciation as nouns and verbs!)

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DATA VISUALISATION

DEGREES OF CHANGE

Adjectives: dramatic, considerable, sharp, significant, substantial

Adverbs: dramatically, considerably, sharply, significantly, substantially

SPEED OF CHANGE

abrupt sudden rapid quick steady gradual slow

PREPOSITIONS

a rise **from** €1m **to** €2m to increase **by** 50%

to fall **by** 30% an increase **of** 7.5 per cent **over** last year

THANK YOU FOR YOUR ATTENTION!

DESCRIBING GRAPHS (VOCABULARY)

UP – verbs:

go up increase take off rise shoot up soar jump improve rocket grow

UP – nouns:

an increase a rise a growth an improvement an upturn a surge an upsurge an upward trend

DOWN – verbs:

go/come down fall fall off drop decline decrease slip plummet shrink

DOWN – nouns:

a fall a decrease a decline a drop a downturn a downward trend

NO CHANGE – verbs:

remain stable level off stay at the same level

remain constant stagnate stabilise

AT THE TOP – verbs:

reach a peak peak top out

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AT THE BOTTOM – verbs:

reach a low point bottom out recover

DEGREES OF CHANGE

Adjectives: dramatic, considerable, sharp, significant, substantial, moderate, slight

Adverbs: dramatically, considerably, sharply, significantly, substantially,

moderarely, slightly

SPEED OF CHANGE

abrupt sudden rapid quick steady gradual slow

PREPOSITIONS

a rise **from** €1m **to** €2m to increase **by** 50%

to fall **by** 30% an increase **of** 7.5 per cent **over** last year

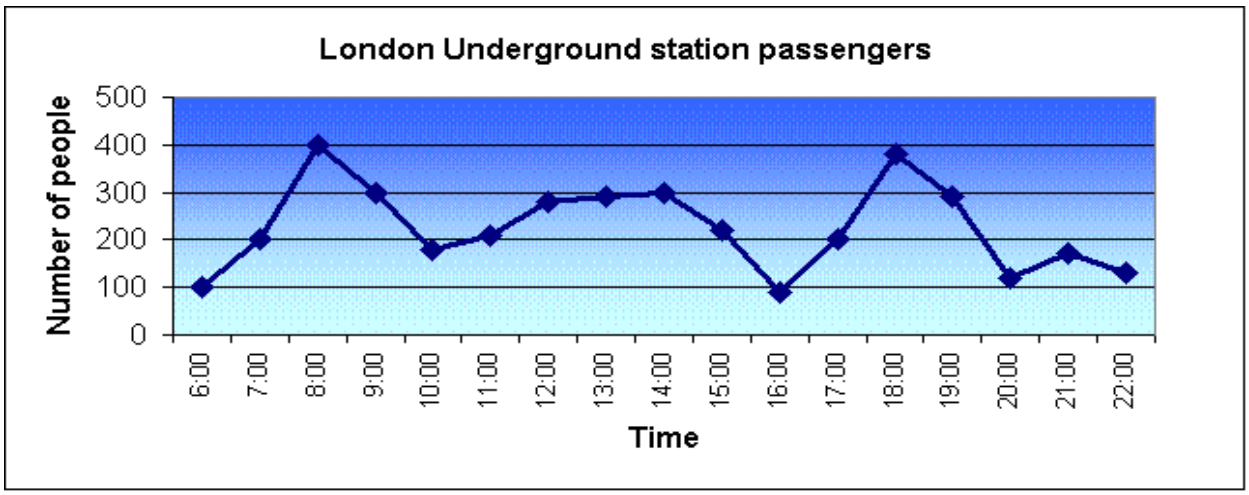
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GRAPH DESCRIPTION

Read the following text then draw a graph based on the description.

The **graph presents** data relating to teenagers and pocket money. A random sample of 1,000 teenagers were surveyed and the average pocket money received at each age has been plotted on the graph. The **x axis** or **horizontal axis** indicates age and the **y axis** or **vertical axis** shows the amount of money received per week. The **graph shows** that 15-year-olds receive twice as much pocket money as 13-year-olds. From the graph we can see that the amount received **reaches a peak** at the age of 18 and then starts to **decline**.

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1. The graph shows that the number of passengers _____ during the day.
2. It is characterized by _____ corresponding to the traveling habits of Londoners..
3. The number of passengers _____ during both the morning and afternoon rush hours.
4. The number _____ at around midday.
5. There was a _____ between 21.00 and 22.00.
6. The number of people on the station _____ between 8.00 and 10.00.
7. The figure _____ again between 14.00 and 16.00.
8. There was, however, another _____ between 16.00 and 18.00.
9. The graph _____ at 16.00.
10. It _____ at 18.00.
11. Nevertheless, the _____ at 8 o'clock in the morning.
12. The number _____ between 6.00 and 8.00.
13. Then, it _____ in just one hour after 8 o'clock.
14. In terms of percentage, it _____ between 10.00 and 13.00
15. The numbers _____ between 17.00 and 18.00.
16. _____ of the day was between 7.00 and 8.00
17. It _____ between 12.00 and 14.00.
18. There was _____ between 19.00 and 20.00.
19. It _____ between 20.00 and 21.00.
20. There was a _____ at 21.00.

all time high was reached	grew only moderately	rose gradually by almost 50%
almost doubled	leveled off	rose sharply
bottomed out	local peak	significant upswing
declined by more than a half	noticeable ups and downs	slight decrease
dropped by a quarter	quadrupled	something of a collapse
fell substantially	reached its afternoon peak	The first dramatic rise
fluctuate considerably	remained relatively stable	

The 5 steps of commentary are demonstrated in these paragraphs:

The graph shows the fluctuation in the number of people at a London underground station over the course of a day.
It is characterized by noticeable ups and downs corresponding to the traveling habits of Londoners.
The busiest time of the day is in the morning. There is a sharp increase between 06:00 and 08:00, with 400 people using the station at 8 o'clock. After this the numbers drop quickly to less than 200 at 10 o'clock. Between 11 am and 3 pm the number rises, with a plateau of just under 300 people using the station. In the afternoon, numbers decline, with less than 100 using the station at 4 pm. There is then a rapid rise to a peak of 380 at 6pm. After 7 pm, numbers fall significantly, with only a slight increase again at 8pm, tailing off after 9 pm.
Comparing the morning and afternoon rush hours, we can notice that they are very similar in both their durations (about 4 hours each) and their peak numbers (about 400). However, the afternoon rush hour reaches a slightly lower peak (just under 400) an falls more dramatically than the morning rush hour.
Overall, the graph shows that the station is most crowded in the early morning and early evening periods.

KEY

1. The graph shows that the number of passengers fluctuate considerably during the day.
2. It is characterized by noticeable ups and downs corresponding to the traveling habits of Londoners..
3. The number of passengers rose sharply during both the morning and afternoon rush hours.
4. The number leveled off at around midday.
5. There was a slight decrease between 21.00 and 22.00.
6. The number of people on the station declined by more than a half between 8.00 and 10.00.
7. The figure fell substantially again between 14.00 and 16.00.
8. There was, however, another significant upswing between 16.00 and 18.00.
9. The graph bottomed out at 16.00.
10. It reached its afternoon peak at 18.00.
11. Nevertheless, the all time high was reached at 8 o'clock in the morning.
12. The number quadrupled between 6.00 and 8.00.
13. Then, it dropped by a quarter in just one hour after 8 o'clock.
14. In terms of percentage, it rose gradually by almost 50% between 10.00 and 13.00
15. The numbers almost doubled between 17.00 and 18.00.
16. The first dramatic rise was between 7.00 and 8.00
17. It remained relatively stable between 12.00 and 14.00.
18. There was something of a collapse between 19.00 and 20.00.
19. It grew only moderately between 20.00 and 21.00.
20. There was a local peak at 21.00.